Completed Gaussian quadratic products: sharp fourth moments and contraction tests

Project Stardust
A companion to a frozen verified library

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Abstract

Centered Gaussian quadratic forms extend from finite matrices to the complete second Wiener chaos. We give complete proofs of the sharp dimension-independent fourth-moment bounds, bounded multiplication into Gaussian L^2, its exact norm, and the degree-zero/two/four product decomposition. A nonzero completed quadratic has strictly larger fourth moment than a Gaussian of the same variance, although the lower universal coefficient is not attained. Polarization gives nonnegative covariance of quadratic squares with exact contraction conditions for equality. For finite symmetric matrices these conditions reduce to matrix annihilation, yielding the classical independence test. The completion and its source/operator normalization are made explicit. Classical quadratic-form theory is credited throughout; no new independence criterion or fourth-moment phenomenon is claimed.

1 The actual Gaussian space and the main statements

Let $(G_i)_{i\ge 0}$ be independent standard real Gaussian variables on their product probability space. Every L^p space below uses this positive probability law. Let $\mathcal{H}_2$ be the closed subspace of L^2 generated by the centered quadratic monomials

$$G_i^2 - 1, \qquad G_iG_j \quad (i<j).$$

This is the second Wiener chaos. Orthogonality of the tensor Hermite polynomials also defines the closed degree-k spaces $\mathcal{H}_k$ and their orthogonal projections P_k. We use the probabilists' normalization $H_2(x) = x^2 - 1$, $H_4(x) = x^4 - 6x^2 + 3$. Here $\mathcal{H}_0$ is the constant space and $P_0F = (\mathbb{E}F)1$.

Theorem 1.1. *Every $F \in \mathcal{H}_2$ belongs to L^4. If $F \neq 0$, then*

$$3(\mathbb{E}F^2)^2 < \mathbb{E}F^4 \le 15(\mathbb{E}F^2)^2. \tag{1}$$

The upper coefficient is attained. The lower coefficient 3 is optimal over the entire space, but equality there forces $F = 0$. Multiplication is a bounded bilinear map

$$(F,G) \longmapsto FG : \mathcal{H}_2 \times \mathcal{H}_2 \to L^2, \qquad \|FG\|_2 \le \sqrt{15}\|F\|_2\|G\|_2,$$

whose norm is exactly $\sqrt{15}$. Its range lies in $\mathcal{H}_0 \oplus \mathcal{H}_2 \oplus \mathcal{H}_4$, rather than in $\mathcal{H}_2$ alone. The exact fourth-cumulant identity is

$$\mathbb{E}F^4 - 3(\mathbb{E}F^2)^2 = \frac{3}{2}\|P_2(F^2)\|_2^2. \tag{2}$$

For all $F, G \in \mathcal{H}_2$,

$$\begin{aligned}\operatorname{Cov}(F^2, G^2) &= 2\langle F, G\rangle^2 + \frac{1}{2}\langle P_2(F^2), P_2(G^2)\rangle + \|P_2(FG)\|_2^2 \\ &\geq 2\langle F, G\rangle^2 \geq 0.\end{aligned} \tag{3}$$

Every term on the first line's right side is nonnegative. In particular equality $\operatorname{Cov}(F^2, G^2) = 0$ holds precisely when all three displayed terms vanish.

The assertion about the middle contraction pairing is special to quadratic chaos. It is not a generic assertion that inner products of arbitrary orthogonal projections are nonnegative. Likewise, uncorrelated quadratic observables need not be independent.

Lancaster's primary quadratic-form work [1] identifies cumulants with matrix traces. Ogawa's primary account [2] states the classical centered Gaussian matrix-annihilation independence criterion. The present paper reconstructs the exact completed-space mechanisms in a verified library; it does not claim priority for those classical results. Primary full-text access for Lancaster's detailed hypotheses remains incomplete.

2 Finite matrices, observable normalization and moments

For a real symmetric $n \times n$ matrix A define

$$Q_A = \sum_{i,j<n} A_{ij}(G_iG_j - \delta_{ij}) = G^T AG - \operatorname{tr} A. \tag{4}$$

If a matrix is not symmetric, its observable equals that of $S(A) = (A + A^T)/2$: transpose the double sum to see that the antisymmetric part vanishes. Thus the source observable identifies matrices differing by a skew-symmetric matrix. We first work on the symmetric representative to keep its norm nondegenerate.

The generating function of independent standard Gaussian coordinates is $\mathbb{E}e^{\langle u,G\rangle} = e^{\|u\|^2/2}$. Expanding it to degree four gives

$$\mathbb{E}G_iG_jG_kG_l = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.$$

Consequently,

$$\mathbb{E}Q_A = 0, \qquad \langle Q_A, Q_B\rangle = 2\sum_{i,j} A_{ij}B_{ij}, \qquad \mathbb{E}Q_A^2 = 2\|A\|_{\mathrm{HS}}^2. \tag{5}$$

The Hilbert–Schmidt norm here is the Frobenius norm on finite matrices, $\|A\|_{\mathrm{HS}}^2 = \sum_{i,j} A_{ij}^2$.

An orthogonal change of coordinates preserves the standard Gaussian law, since it preserves its characteristic function $e^{-\|u\|^2/2}$. Diagonalize A by a real orthogonal matrix. If its eigenvalues are a_i, then Q_A has the law of $\sum_i a_i(G_i^2 - 1)$. Set $U_i = G_i^2 - 1$. The elementary normal moments $\mathbb{E}G^{2k} = (2k-1)!!$ follow by Gaussian integration by parts, and give

$$\mathbb{E}U_i = 0, \qquad \mathbb{E}U_i^2 = 2, \qquad \mathbb{E}U_i^4 = 60.$$

In the fourth power of the independent sum, only the four-equal and two-pair index patterns survive. Thus

$$\begin{aligned}\mathbb{E}Q_A^4 &= 60\sum_i a_i^4 + 24\sum_{i<j} a_i^2 a_j^2 \\ &= 12\Big(\sum_i a_i^2\Big)^2 + 48\sum_i a_i^4.\end{aligned} \tag{6}$$

Since $\sum_i a_i^4 \le (\sum_i a_i^2)^2$, this proves the finite upper bound $\mathbb{E}Q_A^4 \le 15(\mathbb{E}Q_A^2)^2$. Also

$$\mathbb{E}Q_A^4 - 3(\mathbb{E}Q_A^2)^2 = 48\|A^2\|_{\mathrm{HS}}^2, \tag{7}$$

because the eigenvalues of A^2 are a_i^2. No simultaneous diagonalization of two unrelated symmetric matrices is used.

3 The full second chaos and its original source field

Let $\mathcal{S}_2$ be the real Hilbert space of symmetric square-summable matrices $A = (A_{ij})_{i,j\ge 0}$, with its Hilbert–Schmidt norm. Let $A^{(n)}$ retain its first n rows and columns and set all other entries to zero. Then $A^{(n)} \to A$ in that norm. By (5), the actual random variables $Q_{A^{(n)}}$ are Cauchy in L^2. Define

$$Q_A = \lim_{n\to\infty} Q_{A^{(n)}} \quad \text{in } L^2.$$

The resulting map satisfies

$$\|Q_A\|_2^2 = 2\|A\|_{\mathrm{HS}}^2. \tag{8}$$

It is onto $\mathcal{H}_2$: its range is closed by this isometry and contains every centered quadratic monomial. Conversely, every finite-box observable lies in the closed span defining $\mathcal{H}_2$. This proves both inclusions and uniqueness of the symmetric matrix. The definition does not require $\sum_i A_{ii}$ to converge. The formal expression $G^T AG - \operatorname{tr} A$ is only the finite formula; the completed centered quadratic is an L^2 limit, not an unjustified subtraction of two infinite quantities.

The same matrix has an actual Gaussian linear source field. Let $H = \ell^2(\mathbb{N})$ with orthonormal basis (e_i) and define

$$V_A = \lim_{n\to\infty} \sum_{i,j<n} A_{ij} e_i G_j \quad \text{in } L^2(\Omega; H).$$

Orthonormality gives $\|V_A\|^2 = \|A\|_{\mathrm{HS}}^2$. On a finite linear field $V = (V_i)$, the Gaussian divergence is

$$\delta V = \sum_i (G_i V_i - \partial_i V_i).$$

Hence $\delta V_A = Q_{S(A)}$ for finite matrices, and $\|\delta V_A\|_2^2 = 2\|S(A)\|_{\mathrm{HS}}^2$. It extends boundedly to the completion of linear fields, with norm $\sqrt{2}$, and agrees with the original closed divergence there. To check the last assertion, for each smooth cylindrical test f, finite Gaussian integration by parts gives

$$\mathbb{E}(f\,\delta V_{A^{(n)}}) = \mathbb{E}\langle Df, V_{A^{(n)}}\rangle_H.$$

Pass to the two L^2 limits. This is precisely the adjoint-domain identity defining the original divergence, with a unique output. Thus bounded completion is compatible with the original operator, rather than defining a second unrelated divergence.

For symmetric A, the recovery field of Q_A is V_A and has energy $\|V_A\|^2 = \frac{1}{2}\|Q_A\|_2^2$. On finite boxes this is also $\frac{1}{2}DQ_A$, since $DQ_A = 2V_A$. The source's recovery and symmetric-field unitary extend this identity to the full second chaos. In this paper it suffices to retain the explicit coefficient isometry and finite-box approximations. The matrix notation is a human coordinate realization of those source constructions.

4 Strong fourth-moment completion and bounded multiplication

The finite upper bound applies to the symmetric difference of two box matrices. Therefore

$$\|Q_{A^{(n)}} - Q_{A^{(m)}}\|_4 \le 15^{1/4}\|Q_{A^{(n)}} - Q_{A^{(m)}}\|_2.$$

The sequence is Cauchy in L^4. Its L^4 limit has the same L^2 class as Q_A, because the probability measure gives $\|X\|_2 \le \|X\|_4$. Completeness and uniqueness identify the limits. Hence every $F \in \mathcal{H}_2$ belongs to L^4 and

$$\|F\|_4 \le 15^{1/4}\|F\|_2. \tag{9}$$

This also proves the upper bound in (1) on the full space. No continuity of squaring on unrestricted L^2 is used.

By Hölder and (9),

$$\|FG\|_2 \le \|F\|_4\|G\|_4 \le \sqrt{15}\|F\|_2\|G\|_2.$$

The product is the actual almost-everywhere pointwise product, since both factors have L^4 representatives. For box approximations,

$$\|F_nG_n - FG\|_2 \le \|F_n - F\|_4\|G_n\|_4 + \|F\|_4\|G_n - G\|_4 \longrightarrow 0.$$

In particular $F_n^2 \to F^2$ strongly in L^2. Bilinearity on equivalence classes is inherited from pointwise multiplication.

Take $F = G = G_0^2 - 1$. Its variance is 2 and fourth moment is 60. The quotient $\|F^2\|_2/(\|F\|_2\|F\|_2)$ is $\sqrt{60}/2 = \sqrt{15}$. Thus the bilinear norm bound is exact, and the upper fourth-moment coefficient is attained. The image need not be quadratic: $(G_0^2 - 1)^2$ has a nonzero fourth-chaos component, identified below.

5 Degree projections and the completed cumulant

For finite symmetric matrices the degree-two product component is

$$P_2(Q_AQ_B) = 2Q_{AB+BA}, \qquad P_2(Q_A^2) = 4Q_{A^2}. \tag{10}$$

One direct verification uses the degree-two product rule

$$\begin{aligned}(G_iG_j - \delta_{ij})(G_kG_l - \delta_{kl}) &\\ = \text{degree-four Wick term} + \delta_{ik}(G_jG_l - \delta_{jl}) + \delta_{il}(G_jG_k - \delta_{jk}) &\\ + \delta_{jk}(G_iG_l - \delta_{il}) + \delta_{jl}(G_iG_k - \delta_{ik}) + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.&\end{aligned}$$

This identity follows by expanding the multivariate Hermite generating function $\exp(\langle t, G\rangle - \|t\|^2/2)$ in its two pairs of variables. Equivalently, subtract the listed lower degree terms from the product and check its orthogonality to degrees zero through three by Gaussian moments. Summing with coefficients $A_{ij}B_{kl}$ gives (10). It gives $P_0(FG) = \langle F, G\rangle 1$ as well.

For finite observables the product has only degrees $0, 2, 4$. The sum of those three mutually orthogonal closed subspaces is closed: if a sequence in their sum is Cauchy, each projected component is Cauchy and converges in its closed subspace. Strong product convergence therefore gives, for the full second chaos,

$$FG = P_0(FG) + P_2(FG) + P_4(FG). \tag{11}$$

All other degree projections vanish. This is a decomposition in the original Gaussian L^2, not a new multiplication internal to $\mathcal{H}_2$.

For a finite symmetric A, (10) and (5) give

$$\|P_2(Q_A^2)\|_2^2 = 32\|A^2\|_{\mathrm{HS}}^2.$$

Combining this with (7) proves (2) on boxes. Strong L^4 approximation of F and strong L^2 approximation of its square, followed by the bounded orthogonal projection P_2, pass every term to the limit. This proves (2) on the entire second chaos. The argument uses finite-dimensional spectral theory only for the finite fourth-moment calculation; no infinite spectral expansion or trace-class hypothesis on the original quadratic matrix is needed.

Orthogonality in (11) also yields

$$\|P_4(F^2)\|_2^2 = 2\|F\|_2^4 + \frac{1}{2}\|P_2(F^2)\|_2^2. \tag{12}$$

Indeed, $\|F^2\|_2^2 = \|F\|_2^4 + \|P_2(F^2)\|_2^2 + \|P_4(F^2)\|_2^2$, and (2) identifies the left side. In particular,

$$\|P_2(F^2)\|_2^2 \le 8\|F\|_2^4, \qquad 2\|F\|_2^4 \le \|P_4(F^2)\|_2^2 \le 6\|F\|_2^4.$$

These are exact degree-energy consequences, not pointwise positivity statements about the projected functions.

6 Strictness without a uniform gap above three

We prove that the second-degree square contraction of a nonzero completed quadratic cannot vanish. This strictness does not follow merely by taking limits of positive finite cumulants, since positive numbers can tend to zero.

Represent $F = Q_A$ by its unique symmetric square-summable matrix. For fixed i, j, its finite-box contraction satisfies, for $n > i, j$,

$$\frac{1}{4}\big[\text{recovery of } P_2(Q^2_{A^{(n)}})\big]_{ii} = \sum_{k<n} A_{ik}^2 \ge A_{ij}^2.$$

Here recovery sends Q_C to V_C; an individual source coefficient is a bounded linear functional on the completed field. Recovery itself is bounded on the second chaos by the energy identity in Section 3. Since $Q^2_{A^{(n)}} \to F^2$ in L^2, the coefficient inequality passes to the limit. If $P_2(F^2) = 0$, its recovered diagonal coefficient is zero. Thus $A_{ij}^2 = 0$ for every fixed pair i, j, so $A = 0$ and $F = 0$. This proves

$$F \neq 0 \implies P_2(F^2) \neq 0.$$

Equation (2) now gives the strict lower inequality in Theorem 1.1. Since every $F \in \mathcal{H}_2$ is centered, no nonzero such variable has a Gaussian law: a centered Gaussian of variance $\mathbb{E}F^2$ has fourth moment exactly $3(\mathbb{E}F^2)^2$.

Nevertheless there is no larger universal lower coefficient. For $n \ge 1$, set $T_n = \sum_{i=0}^{n-1}(G_i^2 - 1)$. The finite moment calculation gives

$$\mathbb{E}T_n^2 = 2n, \qquad \mathbb{E}T_n^4 = 12n^2 + 48n, \qquad \frac{\mathbb{E}T_n^4}{(\mathbb{E}T_n^2)^2} = 3 + \frac{12}{n} \longrightarrow 3.$$

If a constant c were a lower fourth-moment coefficient for every quadratic in the completed space, these explicit tests would give $c \le 3 + 12/n$ for every n, hence $c \le 3$. Thus strictness for each nonzero target and optimality of the nonattained lower coefficient coexist. No distributional limit theorem for T_n is needed or asserted in this proof.

7 Mixed square covariance by polarization

We next prove the exact mixed identity (3). Set $U = P_2(F^2)$, $V = P_2(G^2)$ and $W = P_2(FG)$. Apply (2) to $F+G$ and $F-G$. Subtract twice its versions for F and G from the sum of those two identities. The fourth-moment contribution is $12\mathbb{E}(F^2G^2)$. Writing $a = \|F\|_2^2$, $b = \|G\|_2^2$, $c = \langle F, G\rangle$, the Gaussian subtraction contributes $-12ab - 24c^2$. On the contraction side,

$$P_2((F \pm G)^2) = U + V \pm 2W.$$

The parallelogram identity then gives the remaining contraction contribution $6\langle U, V\rangle + 12\|W\|_2^2$. Dividing by 12 proves (3). All products and fourth moments exist by Section 4, so polarization is valid on the completed space.

It remains to show $\langle U, V\rangle \geq 0$. For finite symmetric A, B, (10) gives

$$\langle P_2(Q_A^2), P_2(Q_B^2)\rangle = 32\,\mathrm{tr}(A^2B^2) = 32\|AB\|_{\mathrm{HS}}^2 \geq 0. \tag{13}$$

The matrix identity follows from $\|AB\|_{\mathrm{HS}}^2 = \mathrm{tr}((AB)^TAB) = \mathrm{tr}(BA^2B) = \mathrm{tr}(A^2B^2)$, using symmetry and cyclicity of finite traces. Approximate F, G by their finite boxes. Strong square convergence and boundedness of P_2 make the two projected squares converge strongly, so their inner products converge. Nonnegativity therefore passes to arbitrary $F, G \in \mathcal{H}_2$. This proves the lower bounds and the exact equality conditions in Theorem 1.1.

8 Finite-matrix annihilation and the independence boundary

For finite symmetric A, B, the preceding identities specialize to

$$\mathrm{Cov}(Q_A^2, Q_B^2) = 8(\mathrm{tr}\,AB)^2 + 16\|AB\|_{\mathrm{HS}}^2 + 8\|AB + BA\|_{\mathrm{HS}}^2. \tag{14}$$

Here $\langle Q_A, Q_B\rangle = 2\,\mathrm{tr}\,AB$ and $P_2(Q_AQ_B) = 2Q_{AB+BA}$, while (13) gives the middle term. Every term on the right is nonnegative. It follows that

$$\mathrm{Cov}(Q_A^2, Q_B^2) = 0 \quad \Longleftrightarrow \quad AB = 0. \tag{15}$$

The reverse implication also uses $BA = (AB)^T = 0$. If Q_A, Q_B are independent, their squares are independent and their finite fourth moments imply factorization of the product of squares. Equation (15) then forces $AB = 0$. This is the source-mapped necessary independence test.

For completeness, the classical converse is an elementary finite-dimensional corollary, rather than an additional formal declaration asserted in the source map. If $AB = 0$, symmetry gives $BA = 0$ and $\mathrm{ran}\,A \perp \mathrm{ran}\,B$: $\langle Ax, By\rangle = \langle x, ABy\rangle = 0$. Choose an orthonormal basis adapted to these two orthogonal ranges and their joint orthogonal complement. A symmetric matrix annihilates the orthogonal complement of its range; hence Q_A depends only on the Gaussian coordinates in the first range, and Q_B only on those in the second. Those coordinate vectors are independent, so the two observables are independent. Thus (15) is also equivalent to independence in this finite centered Gaussian setting. This is classical quadratic-form theory, not a newly claimed criterion.

Orthogonality of the targets is weaker. For

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$\mathrm{tr}\,AB = 0$, so $\langle Q_A, Q_B\rangle = 0$. But $AB \neq 0$, $AB + BA = 0$, and $\|AB\|_{\mathrm{HS}}^2 = 2$. Equation (14) gives $\mathrm{Cov}(Q_A^2, Q_B^2) = 32 > 0$, so they are dependent. The point is the retained contraction, not covariance of the unsquared targets. This explicit example is a finite matrix substitution in the mapped identities, rather than a new standalone formal theorem.

9 Scope and formal correspondence

All sources are frozen at the verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The theorem map records exact declarations, full headers, source hashes and baseline links. The original full-build and complete transitive-axiom receipt and its referenced evidence hashes are checked separately. No research source was edited and no competing full research build was run.

The principal sources concern original linear-field divergence completion and recovery onto the second chaos, finite Gaussian matrix moments, strong box approximation, actual completed products, degree support, exact cumulants, strict contractions and mixed square covariance. The symmetric square-summable matrix model above is a coordinate reconstruction of their original source-field unitary. The direct L^4 Cauchy proof reorganizes the finite sharp bound; the formal square construction instead first identifies an actual L^2 square through bounded approximation. Both retain the same actual target, original measure and operator normalization.

The paper's general completed statements use the projection conditions in (3). The matrix-annihilation independence statement is presented for finite symmetric matrices. No general infinite-dimensional independence classification, Gaussian approximation rate or nonlinear process identification is silently added. A Wick covariance power alone does not identify a fractional process, generator or stochastic calculus.

Lancaster's primary indexed Theorem 1 gives the classical trace-cumulant antecedent; its full theorem comparison remains incomplete. Ogawa's primary introductory centered theorem and finite orthogonal-range discussion supply the independence context. The formalization value here is the complete extension to original Gaussian L^2 targets and products with sharp constants, together with explicit nonattainment and retained contraction terms. None of these classical mechanisms is represented as a priority claim.

References

[1] H. O. Lancaster, *Traces and cumulants of quadratic forms in normal variables*, Journal of the Royal Statistical Society, Series B **16** (1954), 247–254. doi:10.1111/j.2517-6161.1954.tb00167.x.

[2] J. Ogawa, *On the independence of quadratic forms in a non-central normal system*, Osaka Mathematical Journal **2** (1950), 151–159. doi:10.18910/8824.